INCH-POUND

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SUPERSEDING

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# PERFORMANCE SPECIFICATION

# FILTRATION MODULE, OILY WASTE MEMBRANE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification identifies the performance requirements for a filtration module utilizing a ceramic membrane, which is intended for use in an oily waste membrane system utilizing filtration modules.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-PRF-680 | - | Degreasing Solvent |
| MIL-S-901 | - | Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for |
| MIL-DTL-5624 | - | Turbine Fuel, Aviation, Grades JP-4 and JP-5 |
| MIL-PRF-9000 | - | Lubricating Oil, Shipboard Internal Combustion Engine, High Output Diesel |
| MIL-D-16791 | - | Detergents, General Purpose (Liquid, Nonionic) |
| MIL-PRF-16884 | - | Fuel, Navy Distillate |
| MIL-PRF-17331 | - | Lubricating Oil, Steam Turbine and Gear, Moderate Service |
| MIL-F-24385 | - | Fire Extinguishing Agent, Aqueous Film-Forming Foam (AFFF) Liquid Concentrate, for Fresh and Seawater |

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-167-1 | - | Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited) |
| MIL-STD-740-1 | - | Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment |
| MIL-STD-740-2 | - | Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment |
| MIL-STD-810 | - | Environmental Engineering Considerations and Laboratory Tests |

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

| | | |
|---|---|---|
| NAVSEA DWG 803-6983497 | - | Membrane Module Interface Drawing |

(Copies of this document are available from Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

| | | |
|---|---|---|
| EPA Method 1664 | - | N-Hexane Extractable Material (HEM; Oil and Grease) and Silica Gel Treated N-Hexane Extractable Material (SGT-HEM; Non-Polar Material) by Extraction and Gravimetry |

(Copies of this document are available from Environmental Protection Agency, National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161 or online at www.epa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

| | | |
|---|---|---|
| ISO 12103-1 | - | Road Vehicles -- Test Dust for Filter Evaluation -- Part 1: Arizona Test Dust |

(Copies of these documents are available from International Organization for Standardization (ISO), 1 rue de Varembé, Case postale 56, CH-1211, Geneva 20, Switzerland or online at www.iso.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Description. The filtration module consists of a housing, membrane, piping, mounting clamps and other miscellaneous components. The filtration module interfaces with Navy oily waste membrane systems (OWMS) of any operating capacity utilizing filtration modules to produce permeate water from an oily waste solution.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.



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3.3 Conformance inspection. Each filtration module shall be subjected to conformance inspection in accordance with 4.3.

3.4 Material. The contractor shall select the materials, but the materials selected shall be capable of meeting all of the operational and environmental requirements specified herein. All materials shall be commercially available and conform to applicable commercial marine standards and practices.

3.4.1 Wetted materials. All wetted surfaces shall be compatible, with no evidence of deleterious effect, with the following: seawater; diesel fuel marine fuels specified in MIL-F-16884; turbine fuels specified in MIL-DTL-5624, 2190 TEP steam lube oil specified in MIL-PRF-17331; 9250 diesel lube oil specified in MIL-PRF-9000; contaminants such as bleach, acetone, paint thinner, degreasing solvent specified in MIL-PRF-680 (Type III); aqueous film forming foam specified in MIL-F-24385. In addition, the filtration module housing shall be erosion and corrosion resistant (e.g., glass reinforced plastic, titanium, etc.) and shall not degrade membrane performance.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Sacrificial anodes are not permitted.

3.4.3 Material deterioration, prevention and control. The filtration module shall be fabricated from compatible materials, inherently resistant to or treated to provide protection against corrosion, erosion and microbial deterioration for the unit's service life and in any shipboard and storage environment specified herein.

3.4.4 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent.

3.4.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Environmental considerations.

3.5.1 Shock. The filtration module shall meet the requirements of Grade B, Class I, Type A shock in accordance with MIL-S-901 when fully assembled and supported in accordance with NAVSEA Standard Drawing 803-6983497.

3.5.2 Environmental vibration. The filtration module shall meet the environmental Type I vibration requirements in MIL-STD-167-1 when fully assembled and supported in accordance with NAVSEA Standard Drawing 803-6983497 in a vertically oriented configuration, while pressurized with water or oily concentrate to 100 pounds per square inch gauge (psig).

3.5.3 Airborne noise. The filtration module shall meet the requirements of Airborne Noise in MIL-STD-740-1 for Grade E equipment.

3.5.4 Structureborne noise. The filtration module shall meet the requirements of structureborne noise in MIL-STD-740-2 for Type II equipment.

3.5.5 Operational temperatures (ambient air). The filtration module shall meet specified performance requirements when operating in an ambient air temperature environment range of 50 to 122 degrees Fahrenheit ($^{o}$F).

3.5.6 Storage temperatures (ambient air). When in a non-operating state, the filtration module shall not be damaged nor shall subsequent operational performance be degraded as a result of being subjected to ambient air temperatures ranging from -40 $^{o}$F to 158 $^{o}$F.

3.5.7 Humidity. The filtration module shall not be damaged nor shall subsequent operational performance be degraded when subjected to the combined temperature and relative humidity profiles in MIL-STD-810, Method 507.4.



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3.5.8 Inclination. When fully assembled and supported in accordance with NAVSEA Standard Drawing 803-6983497 in the vertical direction, the filtration module shall operate as specified herein and prevent loss of fluid when subjected to a static inclination angle of 15 degrees from its normal operating position, applied separately in 90 degree intervals about the vertical axis.

3.6 Performance requirements.

3.6.1 Hydraulic resistance. Each filtration module shall impose an axial pressure drop as a function of volumetric flow rate that falls between the maximum and minimum membrane resistance curves (starting at the origin) as shown in Figure 1. The oily waste membrane system will provide hydraulic power as defined by the concentrate flow curve between the respective maximum and minimum curves to the filtration module, also shown in Figure 1. The design operating region is the area bound by the maximum and minimum concentrate flow and the membrane resistance curves.

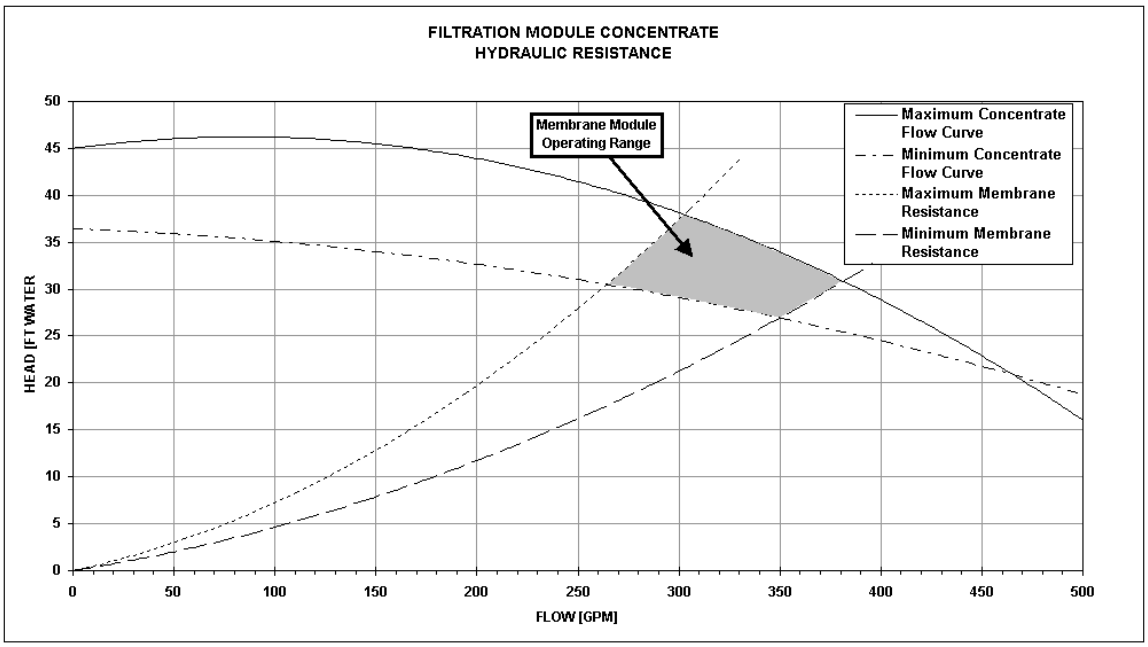



FIGURE 1. Filtration module concentrate hydraulic resistance.

3.6.2 Oil removal. When operating at rated flow and pressure, the filtration module shall deliver permeate water containing no more than 15 parts per million (ppm) of free oil, as measured by EPA 1664 Silica Gel Treated – Hexane Extractable Materials (SGT-HEM), for 95 percent of the time from concentrated fluid containing up to 10,000-ppm oil.

3.6.3 Cross-contamination. The filtration module seals shall not permit cross-contamination from the concentrate side into the permeate system.



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3.6.4 Permeate flow. When processing concentrated oily waste containing up to 10,000 ppm oil and while operating in the defined hydraulic operating region in Figure 1, the filtration module shall deliver a nominal permeate flow rate of 3.3 gallons per minute (gpm). The filtration module shall be able to withstand 105-psig static pressure and 85-psig transmembrane pressure (see 6.7.11).

3.6.5 Operating life. The filtration module shall be capable of operating in compliance with 3.6.1 thru 3.6.4 for at least 300 hours before membrane regeneration.

3.6.6 Hydrostatic integrity. The complete filtration module shall be able to withstand a hydrostatic pressure 1.5 times the filtration module's maximum design pressure specified in 3.7.2 for a period not less than 15 minutes. When pressure is applied and maintained for the specified period, there shall show no sign of leakage, material deformation or rupture, or other defects that harmfully affect the performance and serviceability of the filtration module.

3.6.7 Temperature integrity (fluid). The filtration module shall be able to withstand fluid supply temperatures up to 150 °F at the maximum design pressure specified in 3.7.2 without damaging the unit or degrading performance.

3.6.8 Thermal shock (fluid). When operating at any fluid temperature between 70 °F and the maximum specified in 3.6.7, the filtration module shall be able to withstand the sudden introduction of fluid at a temperature of 70 °F without damaging the unit or degrading performance.

3.6.9 Maintenance. The filtration module shall require no manual preventive maintenance once installed in an oily waste membrane system utilizing filtration modules. When in operation, the membrane shall be capable of being back flushed with potable water at an operating pressure not to exceed 40 psig after every operating cycle at a frequency of approximately every six hours of operation, without damaging the unit nor degrading performance. Performance of this cleaning cycle and regeneration of the membrane shall not require disassembly of the module. In addition, the membrane face shall be easily accessible for reasons such as cleaning after a membrane face plugging (see 6.7.5).

3.7 Interface requirements. The filtration module shall interface with the recirculation system and the permeate discharge systems of an oily waste membrane system utilizing filtration modules.

3.7.1 Physical interface requirements. The filtration module's physical interface requirements shall be in accordance with drawing number NAVSEA Standard Drawing 803-6983497. The filtration module shall be installed in the vertical direction.

3.7.2 Functional interface requirements. The filtration module shall have a design pressure range of 0 to 105 psig. Flow of the permeate water delivered by filtration module shall be controllable by the permeate discharge system of an oily waste membrane system utilizing filtration modules.

3.8 Markings. The filtration module shall have permanently affixed and legible markings that identify the manufacturer's name, model and serial number, and national stock number (NSN), if specified (see 6.2).

3.9 Safety. The filtration module shall present no uncontrolled hazards to operating or maintenance personnel.

3.10 Lifting attachment. A permanently affixed lifting attachment, such as a lifting eye, shall be provided on the filtration module to permit it to be lifted while in its normal operating position.

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).



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4.2 First article inspection. The first article inspection shall be performed on a minimum of one membrane filtration module. This inspection shall consist of the examination of 4.4 and the tests as specified in Table I.

TABLE I. Requirements and verification methods.

| Requirement Title | Requirement Paragraph | 1st Article Test Method | Conformance Test Method |
|---|---|---|---|
| Materials | 3.4 | 4.5.1 | N/A |
| Wetted materials | 3.4.1 | 4.5.1 | N/A |
| Dissimilar metals | 3.4.2 | 4.5.1 | N/A |
| Material deterioration, prevention and control. | 3.4.3 | 4.5.1 | N/A |
| Identification of materials and finishes | 3.4.4 | 4.5.1 | N/A |
| Recycled, recovered, or environmentally preferable materials | 3.4.5 | 4.5.1 | N/A |
| Environmental considerations | 3.5 | N/A | N/A |
| Shock | 3.5.1 | 4.5.5.1 | N/A |
| Environmental vibration | 3.5.2 | 4.5.5.2 | N/A |
| Airborne noise | 3.5.3 | 4.5.5.3 | N/A |
| Structureborne noise | 3.5.4 | 4.5.5.4 | N/A |
| Operational temperatures (ambient air) | 3.5.5 | 4.5.5.5, 4.5.5.6 | N/A |
| Storage temperatures (ambient air) | 3.5.6 | 4.5.5.5, 4.5.5.6 | N/A |
| Humidity | 3.5.7 | 4.5.5.7 | N/A |
| Inclination | 3.5.8 | 4.5.5.8 | N/A |
| Performance requirements | 3.6 | N/A | N/A |
| Hydraulic resistance | 3.6.1 | 4.5.6 | N/A |
| Oil removal | 3.6.2 | 4.5.6 | N/A |
| Cross-contamination | 3.6.3 | 4.5.4 | 4.5.4 |
| Permeate flow | 3.6.4 | 4.5.6 | N/A |
| Operating life | 3.6.5 | 4.5.6 | N/A |
| Hydrostatic integrity | 3.6.6 | 4.5.2 | 4.5.2 |
| Temperature integrity (fluid) | 3.6.7 | 4.5.3 | N/A |
| Thermal shock (fluid) | 3.6.8 | 4.5.3 | N/A |
| Maintenance | 3.6.9 | 4.4, 4.5.2 | 4.5.2 |
| Interface requirements | 3.7 | N/A | N/A |
| Physical interface requirements | 3.7.1 | 4.4 | 4.4 |
| Functional interface requirements | 3.7.2 | 4.4, 4.5.6 | N/A |
| Markings | 3.8 | 4.4 | 4.4 |
| Safety | 3.9 | 4.11 | 4.11 |
| Lifting attachment | 3.10 | 4.4 | 4.4 |



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4.3 Conformance inspection. The conformance inspections shall be random and include not less than ten percent of production units. The conformance inspections shall include the examination of 4.4 and the tests as specified in Table I.

4.4 Examination. Each membrane filtration module shall be examined to determine compliance with 3.7.1, 3.7.2, 3.8 (see Table I) and 3.10. These inspections shall encompass all visual examinations, dimensional measurements, interface requirements, and component and material compliance. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests.

4.5.1 Material test. Conformance to 3.4 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Hydrostatic pressure test. The filtration module shall be subjected to a hydrostatic pressure test as specified in 3.6.6 using potable water supplied through the supply connection at temperatures between 140 °F and 150 °F for first article, and between 65 °F and 85 °F for production units. The hydrostatic test apparatus shall have a combined supply/bleed connection, which shall be connected to either of the filtration module's concentrate interfaces, and to the filtration module's permeate connection. This test apparatus configuration is required in order to prevent damage to the filtration module. The remaining membrane concentrate interface is blanked off for the test. After the test and upon return to atmospheric pressure, the filtration module shall be drained and allowed to air dry. At the conclusion of the hydrostatic test, perform a backflush of the membranes to verify the maintenance requirement in 3.6.9.

4.5.3 Temperature integrity and thermal shock test (fluid). This test shall demonstrate the filtration module's ability to withstand fluid supply temperatures as specified in 3.6.7 and 3.6.8 at a pressure of 100 psig and at a maximum supply flow rate of 2.0 gpm for a period of no less than 15 minutes. This test shall be conducted a total of 20 times on the same module assembly using potable water, 10 times at a water supply temperature of 70 °F, and 10 times at a water temperature of 150 °F. The test apparatus shall have a combined supply/bleed connection, which shall be connected to either of the filtration module's concentrate interfaces, and to the filtration module's permeate connection. The filtration module shall be pressurized with potable water through the supply connection.

4.5.4 Membrane filtration module/housing integrity test. This test shall follow the temperature integrity test in 4.5.3 to demonstrate the membrane-to-housing seal integrity requirement in 3.6.3. The following shall be performed:

a. Using potable water, operate the filtration module in a cross-flow filtration membrane system within the hydraulic design operational region shown in Figure 1 and permeating at 3.7 gpm.

b. Add skim milk to the concentrate as a turbidity marker such that concentrate turbidity exceeds 800 Nephelometric Turbidity Units (NTU).

c. Three permeate samples shall be taken, allowing three minutes between each sample.

d. Concentrate fluid temperature shall not exceed 104 °F.

e. Turbidity of the three samples shall be less than 1.2 NTU. A one to one ratio (half sample, half TCA solution), 24 percent Trichloroacetic Acid (TCA) solution shall be added to each of the samples, agitating for two minutes. The turbidity of each TCA / sample solution, as measured 5 minutes after mixing TCA and sample, shall be less than 20 NTU.

f. Upon completion of the integrity test, the filtration module shall be cleaned to remove skim milk residue using a water solution containing 0.13 grams of Alconox per liter of water and 0.013 liter of Bleach per liter of water.



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4.5.5 Environmental factor verification.

4.5.5.1 Shock test. All components of the membrane filtration module shall be shock tested in accordance with MIL-S-901 as specified in 3.5.1. Since this test is often destructive, use of a nonfunctional membrane is allowable.

4.5.5.2 Environmental vibration test. All components of the first article membrane filtration module shall be tested to MIL-STD-167-1 vibration Type I as specified in 3.5.2.

4.5.5.3 Airborne noise test. The filtration module shall be tested in accordance with MIL-STD-740-1 as specified in 3.5.3.

4.5.5.4 Structureborne noise test. The filtration module shall be tested for structureborne noise tests in accordance with MIL-STD-740-2 as specified as specified in 3.5.4.

4.5.5.5 High temperature. The filtration module shall be tested for high temperature storage and operation in accordance with MIL-STD-810, method 501.4, procedure I and II. The maximum test temperature for operating shall be the upper limit specified in 3.5.5. The maximum test temperature for the storage temperature shall be the upper limit specified in 3.5.6. The operating period for each test shall be one hour at the constant maximum temperature.

4.5.5.6 Low temperature. The filtration module shall be tested for low temperature storage and operation in accordance with MIL-STD-810, method 502.4, procedure I and II. The lowest test temperature for operating shall be the lower limit specified in 3.5.5. The lowest test temperature for the storage temperature shall be lower limit specified in 3.5.6. The operating period for each test shall be one hour at the constant minimum temperature.

4.5.5.7 Humidity. The filtration module shall be tested for the effects of a warm humid environment in accordance with MIL-STD-810, method 507.4.

4.5.5.8 Inclination. The filtration module shall be subjected to the inclination angle specified in 3.4.8. While maintaining the inclination angle, the filtration module shall then be operated for a sufficient period to ensure that continuous operation can be maintained. The filtration module shall then be rotated through the vertical to the specified inclination angle in the opposite direction and the test for continuous operation repeated. The test shall be repeated with the filtration module rotated 90 degrees through the vertical to the plane in which it was originally tested.

4.5.6 Operational test. This test shall demonstrate that the filtration module meets the hydraulic requirements of 3.6.1, effluent performance requirement in 3.6.2, permeate flow and trans membrane pressure requirements in 3.6.4, and the operating life requirement in 3.6.5. The following shall be performed:

a. Operate the filtration module in a cross-flow filtration membrane system at its concentrate hydraulic operational design point while permeating at 3.3 gpm for 300 hours of operation as shown in Table II. The cross-flow filtration membrane system shall concentrate the feed stream by a 100 to 1 volume reduction factor, with the feed stream and concentrate temperatures not exceeding 70 °F and 85 °F, respectively.

b. The filtration module shall process the feed test mixtures described in Table II in the sequential order given, for the designated duration. Each test mixture shall be fed into the system in a manner that insures complete mixing is reached. While processing each test mixture, random permeate and feed samples shall be collected and analyzed in accordance with Table II and 4.5.6.1 to verify the oil removal capabilities in 3.6.2. At least one-quarter of the samples shall be taken during the last one-third of the operational cycle. The concentrate of the membrane system shall be drained from the membrane system after test sequences 1, 2, 3, 4, 6, 8, 10, and 11.

c. During the test, monitor the filtration module hydraulic resistance, membrane-to-housing seal integrity, permeate flow rate, and trans membrane pressure to verify the requirements in 3.6.1, 3.6.3 and 3.6.4 are maintained for the entire 300-hour test duration.



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d. Successful completion of a through c above shall demonstrate compliance of the operating life requirement, without membrane regeneration, in 3.6.5.

TABLE II. Operational test parameters.

| Seq. | Hours | Feed Test Mixture | Number of Feed Samples | Number of Permeate Samples |
|---|---|---|---|---|
| 1 | 60 | 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2], and 50 ppm ISO 12103-1 fine test dust in potable water | 2 | 11 |
| 2 | 60 | 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2], and 50 ppm ISO 12103-1 fine test dust in potable water | 2 | 11 |
| 3 | 60 | 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2], and 50 ppm ISO 12103-1 fine test dust in potable water | 2 | 12 |
| 4 | 6 | 0.5% bleach and 100 ppm Navy oil mix #4, 25 ppm detergent mix #4[2] in potable water | 0 | 1 |
| 5 | 24 | 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2], and 50 ppm ISO 12103-1 fine test dust in potable water | 1 | 3 |
| 6 | 6 | 0.5% acetone and 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2] in potable water | 0 | 1 |
| 7 | 24 | 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2], and 50 ppm ISO 12103-1 fine test dust in potable water | 1 | 3 |
| 8 | 6 | 0.5% paint thinner and 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2] in potable water | 0 | 1 |
| 9 | 24 | 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2], and 50 ppm ISO 12103-1 fine test dust in potable water | 1 | 3 |
| 10 | 6 | 3% sea salt and 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2] in potable water | 0 | 1 |
| 11 | 24 | 100 ppm Navy oil mix #4[1], 25 ppm detergent mix #4[2], and 50 ppm ISO 12103-1 fine test dust in potable water | 1 | 3 |

Notes:

(1) Navy Oil Mix #4 consists by percent weight of 50% MIL-PRF-16884, 25% MIL-PRF-17331, and 25% MIL-PRF-9000.

(2) Detergent mixture consists by percent weight of 50% MIL-D-16791 (Type I), 25% commercial liquid laundry detergent (i.e., liquid Tide, etc.) and 25% MIL-PRF-680 (Type III).



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4.5.6.1 Oil sampling. The samples shall be collected and analyzed in accordance with EPA Method 1664 SGT-HEM (see Table II). A minimum of three quality control (QC) standard samples containing 10 ppm Navy oil mix #4 and three QC standard samples containing 100 ppm Navy oil mix #4 (prepared by an independent laboratory) shall be inserted randomly into the analysis matrix to evaluate the analysis laboratory's accuracy.

4.5.7 Safety. The filtration module shall be evaluated throughout testing for safety requirements in 3.9.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The oily waste filtration module described in this specification is intended for use aboard vessels in the armed forces in a membrane system to process effluent from an oil/water separator (OWS), reliably producing an effluent containing less than 15 ppm oil content. The overall intended service life of the filtration module is 1,200 hours minimum with regeneration.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of the specification.

b. First article requirement (see 4.2).

c. Conformance requirement (see 4.3).

d. Material certificates (see 4.5.1).

e. Packaging requirements (see 5.1).

6.3 Filtration module. Information regarding the Navy studies in the field of Ultrafiltration may be found on the Internet Website. URL's will be provided by the procuring activity upon request. Developmental information originated from the Naval Surface Warfare Center, Code 634, Bethesda, MD. The filtration module should incorporate an open system architecture strategy as defined in 6.7.7. Also, the center attachment bolt can be shared between two inline filtration modules.

6.4 Recommended test material. Test conducted at Naval Surface Warfare Center used Liquid Tide™ as the commercial laundry detergent.

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been authorized and assigned an Acquisition Management Systems Control (AMSC) number must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.6 Design and fabrication guidelines. Any component design guidelines referenced in this section are based upon past Navy shipboard and laboratory experience with filtration modules. Modules designed under these guidelines, have successfully met the performance requirements specified herein. These guidelines are intended to assist the contractor in ensuring that the filtration module fully meets these performance requirements stated in 3.7.



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6.7 Definitions.

6.7.1 Concentrate. Contaminant-laden fluid that has not passed through the membrane as permeate, also known as retentate.

6.7.2 Corrective maintenance. All actions performed as a result of failure, to restore an item to a specified condition. Corrective maintenance can include any or all of the following steps: localization, isolation, disassembly, interchange, re-assembly, alignment, and checkout.

6.7.3 Cross-flow. As applied to a membrane, defines concentrate flow running parallel to the membrane surface, used to minimize membrane fouling.

6.7.4 Feed. The influent stream to be processed by the filtration modules in the oily waste membrane ultrafiltration system.

6.7.5 Membrane face plugging. The accumulation of solids on the membrane face resulting in conditions outside performance requirements specified in 3.6.4. An example of a clean membrane face free of solids is shown in Figure 2.



FIGURE 2. Clean membrane face.

6.7.6 Oil/water separator (OWS). Generally used to refer to any device, which separates oil and water. The OWS processes oily waste before the oily waste membrane ultrafiltration system.

6.7.7 Open system strategy. A design approach resulting in an open system design that optimizes predicted life cycle cost and performance, allows advances in technology to be readily incorporated, and provide for rapid reconfiguration of systems to respond to mission or operational requirement changes. The open system design favors modularity and compatibility with other Fleet assets and uses standard and/or publicly available, nonproprietary interfaces such that new or upgraded functionality can be incorporated from multiple suppliers over the ship lifecycle. The open system design facilitates operations in combined missions with other services, and interagency, multinational, and non-Government organizations.

6.7.8 Permeate. Water from the concentrate loop that has passed through the membrane, becoming dischargeable water (15 ppm oil, maximum).

6.7.9 Preventive maintenance. All actions performed in an attempt to retain an item in specified condition by providing systematic inspection, detection, and prevention of incipient failures.



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6.7.10 Recirculation system. The part of a complete membrane system that continuously moves the concentrate through the filtration modules providing the cross-flow velocity necessary to minimize membrane fouling.

6.7.11 Transmembrane pressure. The average pressure in then filtration module minus the permeate pressure.

6.8 Subject term (key word) listing.

Bilge water

Filtration device

Membrane

Oil/water separator

Open system architecture

OSA

OWS

Recirculation system

Ultrafiltration

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – AT
Navy – SH
Air Force – 99
DLA – CC

Preparing Activity:
Navy – SH
(Project 4610-0036)

Review Activities:
Air Force – 03, 84

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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